

August 16, 2026 08:07 PM GMT

SpaceX | North America

Blades & Vanes: SpaceX Tackles Time to Power

SpaceX has identified one of the biggest supply chain bottlenecks in terrestrial AI and is set to apply its brand of extreme vertical integration to alleviate the problem. We take a closer look at SpaceX's foundry in Bastrop, Texas and what it means for future SpaceXAI datacenter scale.

If a Datacenter is built in the woods and there's no power to run it... did it compute? SpaceX's [job postings page](#) offers some valuable clues to the company's strategic direction. In the world of converting energy into intelligence at Giga-scale, unlocking the bottlenecks of power is a major factor. There are currently 4 postings on SpaceX's career page for positions related to the Blades and Vanes Foundry in Bastrop, Texas (Automation Engineer, Sr. Automation Engineer, Sr. Operations Engineer, Sr. Materials Engineer).

Intelligence-per-watt-per-dollar-per-second: $((i/w)/\$)/t$. This quotient measures the energy density, capital density and temporal density of intelligence. Imagine you had 3 companies with identical intelligence-per-watt-per-dollar. All else equal, the one that could deliver their $((i/w)/\$)$ in the least amount of time would create the most value. Can SpaceX achieve superior time to convert 'e to i' vs. its peers with the least amount of capital deployed? Drivers include vertical integration amplified by pervasive physical and digital networks.

In the race for 'AI Supremacy' those who can create efficient intelligence for the lowest price at the right place at the right time - persistently, redundantly, reliably (and at scale) will be advantaged.

SpaceX has a history of attacking whatever constrains scale most. As the company builds its moat in compute time and cost-to-power, it appears to have found its next target: turbine blades and vanes. We can identify 3-4 GW of likely accessible gas turbine power by YE27 through a mix of purchase agreements and Musk's personal acquisition of mobile-power provider APR Energy, with total reported commitments closer to 8 GW under any timeline, and likely more that has not been publicly disclosed. Given the scale of its terrestrial compute ambitions, we would expect that number to keep climbing (we see terrestrial compute peaking at ~15 GW in 2031, with upside if orbital compute is delayed). However, the practical constraint on adding more sits in the hot-section blades and vanes inside each machine, a market controlled by a handful of casters already backlogged for years. Job postings point to a greenfield foundry in Bastrop, TX aimed at solving that for both datacenter turbines and Raptor turbopumps, potentially opening the door to partnerships with existing turbine manufacturers for SpaceX's own longer-term industrial gas turbine supply. Although, ramping production may take multiple years given the process complexity, equipment lead times, and workforce build-out

MORGAN STANLEY & CO. LLC

Adam Jonas, CFA

Equity Analyst

Adam.Jonas@morganstanley.com

+1 212 761-1726

William Tackett, CFA

Research Associate

William.Tackett@morganstanley.com

+1 212 761-6028

Kristine T Liwag

Equity Analyst

Kristine.Liwag@morganstanley.com

+1 212 761-2980

David Arcaro, CFA

Equity Analyst

David.Arcaro@morganstanley.com

+1 212 761-1817

Justin M Lang

Equity Analyst

Justin.Lang@morganstanley.com

+1 212 761-6251

Jeremy Herring

Research Associate

Jeremy.Herring@morganstanley.com

+1 212 761-4182

SpaceX (SPCX.O, SPCX US)

Space Technology | United States of America

Stock Rating	Overweight
Industry View	Attractive
Price target	\$300.00
Shr price, close (Aug 13, 2026)	\$141.29
52-Week Range	\$225.64-104.83

Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (\$) **	(1.69)	0.54	2.45	6.36
Prior EPS (\$) **	-	-	-	-
P/E	NM	261.4	57.7	22.2
EPS (\$) §	-	(0.05)	1.73	4.97
Div yld (%)	-	0.0	0.0	0.0

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework

** = Based on consensus methodology

§ = Consensus data is provided by Refinitiv Estimates

e = Morgan Stanley Research estimates

QUARTERLY EPS (\$)

Quarter	2025	2026e Prior	2026e Current	2027e Prior	2027e Current
Q1	(0.18)	-	(0.84)a	-	0.59
Q2	(0.34)	-	(0.09)a	-	0.56
Q3	(0.36)	-	0.21	-	0.62
Q4	(0.80)	-	0.55	-	0.67

e = Morgan Stanley Research estimates, a = Actual Company reported data

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

involved in single crystal casting.

- **Why should investors care?**

- After the exclusivity agreement with NVIDIA, we would think chip supply for SpaceX is mostly secured, at least up to the 10 GW given by Elon Musk on the 2Q call.
- However, both lack of clarity in access to off-grid power and aggressive time-to-deploy is the reason we model just 5 GW by YE27 vs. the 10 GW suggested by Musk. Our SpaceXAI estimates beyond 2027 will depend on ability to tackle this critical bottleneck.
- Securing large-scale turbine access is incredibly important for SpaceX's chances to scale compute to compete more directly with hyperscaler and large AI lab peers. Industrial gas turbines for terrestrial compute + turbopumps for Raptor Engines for orbital compute (>50% of launches beginning in 2029).
- Each GW of SpaceXAI compute represents an opportunity of upwards of \$50bn in annual revenue.
- SpaceXAI's moat = $f(\text{Intelligence/Watt/Dollar/Second})$. Vertically integrating a bottleneck reduces both cost and time-to-deploy which leads to a faster flywheel that improves intelligence per unit of energy.

What to do with the stock? Backing out the value of Space + Connectivity (\$127/share) from the current share price (\$140) implies just \$13 per share for Consumer + Enterprise AI, roughly 1x 2028 EV/Sales and below even neocloud peers. **We expect investors will learn much more on SpaceX's path to securing the power needed to make Elon Musk's 10GW terrestrial compute goal more plausible (MS is at 5GW by YE27). In our view, the lock-up expiry overhang offers an attractive entry-point for a generational compounder converting energy into intelligence at scale. Overweight, PT \$300, bull case \$600.**

Key color from CEO Elon Musk from his February interview with Dwarkesh Patel and John Collison on the Cheeky Pint podcast:

- Dwarkesh Patel: "What is the reason to think those challenges [of orbital datacenters] which have never had to be addressed before will end up being easier than just building more turbines on Earth? There are companies that build turbines on Earth. They can make more turbines, right?"
- Elon Musk: "Again, try doing it and then you'll see. The turbines are sold out through 2030."
- John Collison: "Have you guys considered making your own?"
- Elon Musk: **"In order to bring enough power online, I think SpaceX and Tesla will probably have to make the turbine blades, the vanes and blades, internally."**
- John Collison: "But just the blades or the turbines?"
- Elon Musk: **"The limiting factor... you can get everything except the blades. [...] The limiting factor is the vanes and blades. There are only three casting companies in the world that make these, and they're massively backlogged."**

Why would SpaceX want to in-source key gas turbine supply? There is precedent

for Elon Musk's willingness to vertically integrate to alleviate supply bottlenecks driven by a number of considerations. To be clear, we do not think this is principally about whether megawatts are available. Beyond 2030, there is very possibly adequate turbine capacity, and smaller equipment (aeroderivatives, reciprocating engines, fuel cells) is available even earlier. The argument for in-sourcing is also optionality, flexibility, and cost, i.e. the ability to add power on SpaceX's own schedule rather than an OEM's slot calendar, to co-design the airfoil around the machine and the site, and to capture economics in a value chain where pricing has tripled and OEMs are now recognizing mid-30% EBITDA margins on new orders.

- **Optionality over schedule - "control our long term destiny".** Turbine order books now stretch to the end of the decade. Even where megawatts can be procured, they arrive on the supplier's timeline, in the supplier's configuration.
- **Improvement in end-product design and function - "we can make a better product".** First-principles process control (the language in SpaceX's own job posts) is aimed at improving yield, performance, and cost.
- **Design freedom - "co-design the blade with the machine".** Alloy, cooling scheme, and turbine architecture get developed as one problem rather than sequenced across a supplier interface.
- **Internal dual-use scale - "one foundry, two customers".** Raptor turbopump castings and datacenter turbine airfoils draw on the same alloys, furnaces, and trained workforce, letting a single foundry spread fixed cost across the space and AI businesses.

SpaceX needs a lot of blades and vanes. Both the SpaceX space and AI businesses consume the same difficult-to-produce hot section castings:

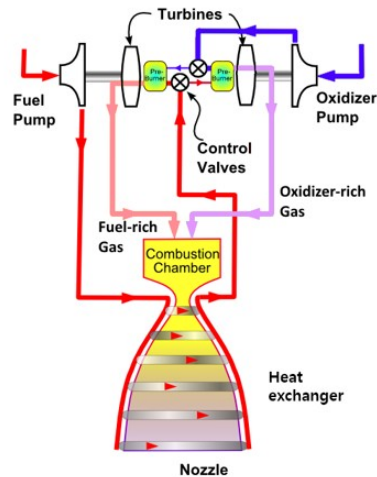
1. **Natural gas turbines for terrestrial AI datacenters:** Per its S-1, SpaceX has committed roughly \$2.8bn to AI-infrastructure turbines: an \$805mn purchase agreement through 2029 (March 2026) plus a ~\$2.0bn mobile gas turbine "Turbine Acquisition" to power its data centers. That second deal potentially corresponds to Musk's acquisition of mobile-power specialist APR Energy, which an [FTC notice dated May 14, 2026](#) named him personally as buyer of, bringing a mobile fleet of over [1.1 GW](#). Every one of these turbines is gated by the same scarce vanes and blades the Bastrop foundry targets.
2. **Turbopumps for Starship Raptor engines:** SpaceX's Raptor is a full-flow staged-combustion engine in which an oxygen-rich turbine drives the oxygen turbopump and a fuel-rich turbine drives the methane turbopump, with turbomachinery already cast from an [Inconel-like custom "SX500" superalloy](#), the same class of nickel-superalloy, single-crystal/directionally-solidified casting the Bastrop foundry is chartered to produce. *We note that it is unclear whether SpaceX currently casts the Raptor turbopump blades or if there are any other external suppliers.* Each V3 Starship requires 39 Raptor engines (33 for booster, 6 for ship)- with SpaceX likely to eventually require [hundreds or even thousands](#) a year.

Exhibit 1: Cross Section of a Mobile Natural Gas Turbine, Showing Turbine Blades



Source: Solar Turbines (subsidiary of Caterpillar)

Exhibit 2: Full Flow Staged Combustion Engine Diagram (Similar Engine Design as Starship's Raptor) Showing Turbines Used to Pump LOX and Liquid Methane to the Combustion Chamber.



Source: Wikipedia (exact author unknown)

Little has been formally announced, but we can piece together details from job postings. Most details on the Bastrop, TX "Blades and Vanes Foundry" are from roles posted back in April (see: [materials](#), [automation](#), and [operations engineers](#)) that describe a greenfield, fully in-house vertically integrated casting operation (the operations role seeks the "extreme owner of the construction, buildout, and operational ramp-up" of a new foundry). The foundry would produce the equiaxed, directionally solidified, and single crystal nickel superalloy castings that for both Raptor engines and the industrial gas turbines for its datacenters.

What does the blades and vanes market look like today? The market is remarkably concentrated. A handful of independent merchant casters dominate the hot-section blade and vane supply: Precision Castparts (Private), Consolidated Precision Products (Private), Howmet Aerospace (HWM) and Doncasters (DPC), the latter two covered by Kristine Liwag. Others participate around the edges while many of the major engine OEMs also cast single crystal airfoils in-house for their own turbines (although a large % still sourced from casters). But the practical reality is a handful of specialists, each backlogged for years, sitting on a highly specialized process that has become the true chokepoint on scaling gas power. EBITDA margins for blades and vanes can be ~15-20%+ (based on DPC which is the cleanest public pure play today).

Single crystal casting is widely described as the most technologically demanding casting process in the world. It may be 2-3+ years before the first volume production for SpaceX.

- The process for creating turbine blades ([a modified Bridgman-Stockbarger process inside a vacuum furnace](#)) requires simultaneous control of dozens of variables (casting temperature, crystal seeding angles, metal flow rates,

viscosity, orifice sizes, differential cooling, etc.) across a 40-50 step manufacturing sequence.

- The alloys involved must survive temperatures exceeding 3,000 degrees Fahrenheit or nearly 800 degrees above their own melting point while spinning at 12,000-15,000 RPM under pressure ratios of 40:1. The single crystal structure (a blade grown from a single grain with no internal boundaries) is the only known solution to surviving these conditions at scale.
- Making a single blade in a lab can be relatively straightforward, but making thousands per week at acceptable yields is a precise art that generally requiring decades of accumulated process know-how that cannot be easily recreated, large-scale and highly-specialized capital equipment with lead times of up to three years, and a large trained workforce.
- Vertical integration is a major prerequisite: the established players control their own superalloy supply, shell systems, coatings, and furnace designs in-house.
- On top of the process complexity, for aerospace, every part must clear stringent OEM qualification and NADCAP certification, approvals that themselves take years to obtain and lock in suppliers at the part-number level, making switching operationally dangerous for engine makers.
- The result is that from a greenfield commitment to first qualified production, the minimum timeline is two years, and full run-rate output typically takes three or more, which would potentially mean any new entrant, including SpaceX, faces a multi-year gap between breaking ground and delivering certified parts at volume.
- Sources: *Howmet 2026 Investor Day and 2Q26 Earnings Call, Doncasters S-1*

Exhibit 3: Current Competitive Landscape for Gas Turbine Blades & Vanes



Source: Morgan Stanley Research

Where is SpaceX sourcing turbines currently, and how much is under contract?

Based on various reports, SpaceX may have sourced from a number of different suppliers potentially including, but not publicly confirmed to be, [Doosan Enerbility](#) (covered by Heewon Choi), Caterpillar Solar Turbines via [Solaris Energy Infrastructure](#) (SEI, covered by David Arcaro), [FTAI](#) (FTAI, covered by Kristine Liwag), [APR Energy](#) - recently personally acquired by Elon Musk - and potentially others. While we have no knowledge of any specific transaction, piecing together reports, we can potentially identify 3-4 GW of power that SpaceX could reasonably access by YE27, possibly with more undisclosed. Adding up all turbine agreements reported to be associated with SpaceX under any timeline would come closer to 8 GW. Given the

scale of SpaceXAI's terrestrial compute ambition and the potential multi-year lead time to produce turbine blades & vanes at scale, we would still expect the number of turbines under contract by SpaceX to continue rise for the foreseeable future. *Note that SpaceX has not commented directly on any of the aforementioned reports.*

Could SpaceX partner with turbine providers to supply blades and vanes? Yes - for internal SpaceX supply. While we can't pinpoint specific potential partners, hot-section castings are the chokepoint for the turbine OEMs themselves, and incremental merchant capacity is additive in a market where incumbents are backlogged for years and AI-driven demand has outrun casting capacity additions. However, we would not expect SpaceX to be a material supplier of blades & vanes for supply outside of SpaceX use.

Exhibit 4: Reported (Not Confirmed) Turbine Supply for SpaceX

Potential Supplier	MW	Type	Confirmed?	Details	Timing	Source
 DOOSAN	4,560	Permanent Turbines	Elon Musk confirmed purchase of initial 5 on X. Additional 7 reported by local media, but not confirmed.	Reported 12x 380 MW permanent turbines	First 2 units due by end-2026; remainder thereafter	SemiAnalysis (Jan 2026), Yonhap (March 2026)
 APR ENERGY	1,100	Mobile Gas Turbines	Acquisition by Elon Musk (personally) confirmed by public FTC filings. SpaceX connection inferred, but not confirmed by company or Elon Musk.	Fleet of 1.1GW of MGT's per APR website. Presumably relates to \$2bn "Turbine Acquisition" disclosed in 8-K	FTC early termination notice dated May 14, 2026; deal closed	FTC
 SOLARIS	1,140	Mobile Gas Turbines	Widely reported. No confirmed, but public images show Solaris turbines at Colossus	Reported 900 MW JV w/ xAI + 240 MW for Colossus 1	Reported 240 MW installed for Colossus 1; new 900 MW phased 2026-2029	Bloomberg, Company, SemiAnalysis
 FTAI AVIATION	> 1,000	Mod-1 CFM56 Aerodrive Units	No. FTAI names only "a leading international cloud service provider" and has not disclosed the customer. SpaceXAI link is inferred, not stated.	\$1.465bn Order with unidentified "International Cloud Service Provider" (Assume \$1-1.5/watt)	Order signed Jul 22, 2026; deliveries in batches through Nov 2027	Company, FundaAI

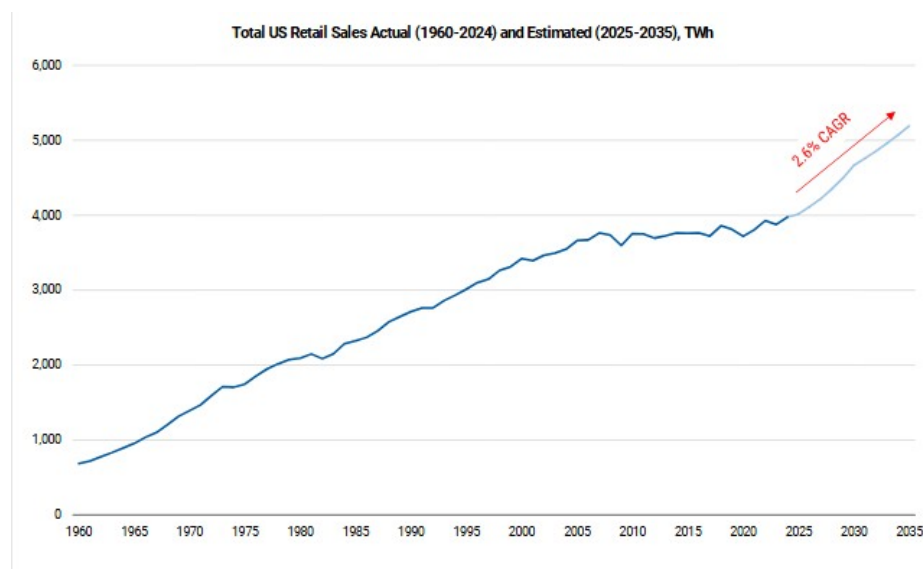
Source: FTC, Bloomberg, Company Filings, SemiAnalysis, Morgan Stanley Research

Please see here for our SpaceX Initiation: SpaceX: AI's Final Frontier; Initiate at Overweight, PT \$300 (7 Jul 2026)

Industrial Gas Turbine Market Overview

Secular Gas Turbine Trends from AI. The gas turbine market has inflected to support growing power demand, largely from powering AI. In the US, the largest market for turbines, we are forecasting power demand will grow ~2.6% annually through 2035 (vs. just ~0.5% annually the past two decades) to support this growth in data centers, reshoring, and electrification. Globally, data center power demand is set to 4x between 2023 and 2028 from ~55 GW to 210 GW. That demand flows directly through to the hot-section supply chain, as every frame shipped by the major power generation OEMs (GEV, ENR, Doosan, and Ansaldo) is gated by vanes and blades sourced from a small set of merchant casters and captive in-house lines. Turbine pricing has increased by as much as 3x to \$3,000/kW and OEMs are set to recognize mid-30% EBITDA margin on newly placed OE turbine orders, compared to 0-5% historically. To that end, turbine backlogs/lead times have also increased and now extend out ~3-5 years with next available delivery slots for most manufacturers being 2029 or 2030. Put together, the urgency of power demand has forced OEMs and end buyers alike to lock in upstream casting supply years in advance, the same constraint Musk described when he said turbines are sold out through 2030.

Exhibit 5: Meaningful inflection ahead in US power demand

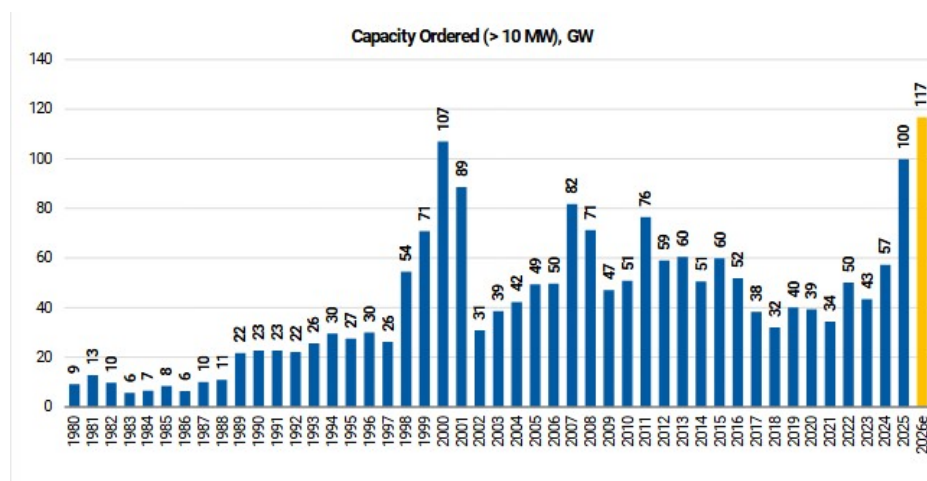


Source: U.S. Energy Information Administration (EIA), Morgan Stanley Research estimates

Buyers are paying up, and paying early. Driven by a need for firm baseload power for data centers, of which our thematic team estimates 150 GW of new US data center demand through 2030, customers placed 100 GW of gas turbine orders in 2025, the second largest year of orders ever. Annualizing 1Q bookings, in 2026 the industry will recognize the all-time largest number of orders (117 GW), across all frame sizes. We believe the cycle has more room to run as utilities and developers have shown an insensitivity to price increases and an interest in signing multi-year 2030 frame agreements to lock in slots to support its build out of generation. To that end, GE Vernova (GEV) has noted that they have yet to reach a price ceiling for turbines, with it now charging customers ~\$3,000/kW for heavy duty CCGTs (vs ~\$1,000/kW for plants

commissioned in the 1H of the 2020s) and ~25% down payments for slot reservations. These data points underpin the durability of the market and, in our view, reinforce the growth outlook for the entire IGT value chain, including the airfoil casters sitting at its narrowest point.

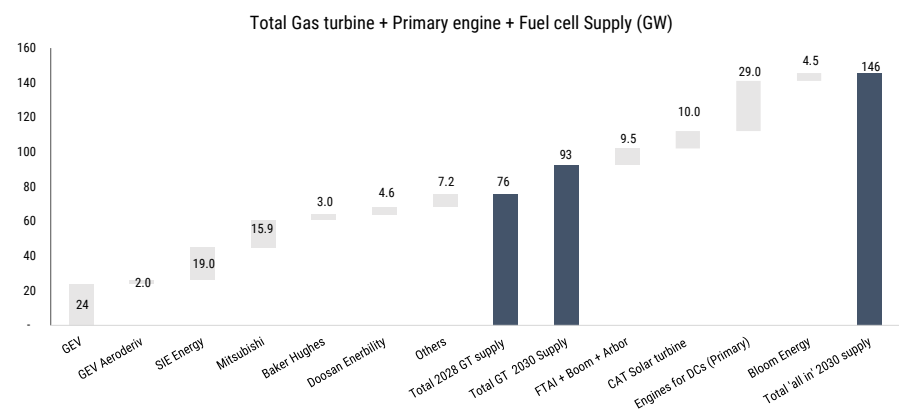
Exhibit 6: Gas turbine order - 2025 full year order intake landed at 100GW (single cycle), which is the second highest order level in history (since 2000). We expect a record year in order intake in 2026, and 1Q26 is annualizing at 117 GW



Source: McCoy, Morgan Stanley Research estimates

Supply is responding, but not evenly. In parallel with the rising demand, supply is also increasing materially from both legacy suppliers and new entrants to help address a potential power shortfall and capture share of the market. From a baseline of ~50-60 GW in 2025, the turbine industry will nearly double to ~97 GW of nameplate capacity by 2030, before considering ~35 GW of alternative engine and fuel cell supply. Casters are supporting that expansion through new long-term supply agreements with key IGT customers, though adding qualified airfoil capacity is slower and harder than adding frame assembly capacity. While oversupply remains a concern, we expect legacy suppliers with heavy-duty frame options will be the most derisked solution longer term given they have the lowest levelized cost of electricity (LCOE), a major consideration for cost conscious utilities. Furthermore, in the near term, online hyperscalers have become creative and are now purchasing/deploying smaller, less economic turbines that historically did not provide 24/7 baseload power (i.e., aeroderivative and industrial gas turbines), the same class of asset behind SpaceX's mobile turbine acquisitions. These smaller frames consume the same superalloy castings as heavy duty units, so both near- and long-term demand trends converge on the same bottleneck.

Exhibit 7: Annual Global Baseload Gas Manufacturing Capacity



Source: Company Filings, Morgan Stanley Research estimates

Valuation Methodology and Risks

SpaceX (SPCX.O)

Sum-of-the-Parts by Segment: Space, Connectivity, AI (Divided into X & Grok and Enterprise AI)

- \$300 Price Target = sum of Space (\$8), Connectivity (\$118), X & Grok (\$8), Enterprise AI (\$165)

2040 ending forecast period

6/30/2027 valuation date

11.1% WACC; 11.9% Cost of Equity.

50% Enterprise AI valuation discount for execution risk

TGR of: 4.0% Space, 4.5% connectivity, 3.0% X & Grok, 5.0% Enterprise AI

Equates to 0.46 EV/EBIT/Growth

Risks to Upside

- Faster Starship reuse progress
- Faster Starlink capacity growth
- Stronger DTC / enterprise adoption
- More neocloud wins
- Cursor ARR acceleration
- Lower AI infrastructure time to power and cost

Risks to Downside

- Slower Starship reuse cadence
- Slower Starlink subscriber growth
- Weaker enterprise AI monetization
- Higher capex / cost per watt of compute
- Longer time-to-power
- Greater funding needs / dilution
- Regulatory delays

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared by Morgan Stanley & Co. LLC, and/or Morgan Stanley C.T.V.M. S.A., and/or Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., and/or Morgan Stanley Canada Limited. As used in this disclosure section, "Morgan Stanley" includes Morgan Stanley & Co. LLC, Morgan Stanley C.T.V.M. S.A., Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., Morgan Stanley Canada Limited and their affiliates as necessary.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: David Arcaro, CFA; Adam Jonas, CFA; Justin M Lang; Kristine T Liwag.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

As of July 31, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: Firefly Aerospace Inc, Iridium Communications Inc, MDA Space Ltd, Planet Labs PBC, Rocket Lab USA Inc, Viasat Inc, Virgin Galactic Holdings Inc, Voyager Technologies Inc.

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Firefly Aerospace Inc, HawkEye 360 Inc, **SpaceX**, Voyager Technologies Inc.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Firefly Aerospace Inc, HawkEye 360 Inc, **SpaceX**, Voyager Technologies Inc.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from Firefly Aerospace Inc, Gogo Inc, HawkEye 360 Inc, Iridium Communications Inc, MDA Space Ltd, Planet Labs PBC, Rocket Lab USA Inc, **SpaceX**, Viasat Inc, Voyager Technologies Inc.

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from Gogo Inc, HawkEye 360 Inc, Iridium Communications Inc, Rocket Lab USA Inc, **SpaceX**, Viasat Inc, Voyager Technologies Inc.

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Firefly Aerospace Inc, Gogo Inc, HawkEye 360 Inc, Iridium Communications Inc, MDA Space Ltd, Planet Labs PBC, Rocket Lab USA Inc, **SpaceX**, Viasat Inc, Voyager Technologies Inc.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: Gogo Inc, HawkEye 360 Inc, Iridium Communications Inc, Rocket Lab USA Inc, **SpaceX**, Viasat Inc, Virgin Galactic Holdings Inc, Voyager Technologies Inc.

Morgan Stanley & Co. LLC makes a market in the securities of Iridium Communications Inc, Viasat Inc.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of July 31, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Coverage Universe			Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1529	42%	442	47%	29%	750	43%
Equal-weight/Hold	1582	43%	396	42%	25%	773	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	560	15%	97	10%	17%	217	12%
Total	3,674		936			1741	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

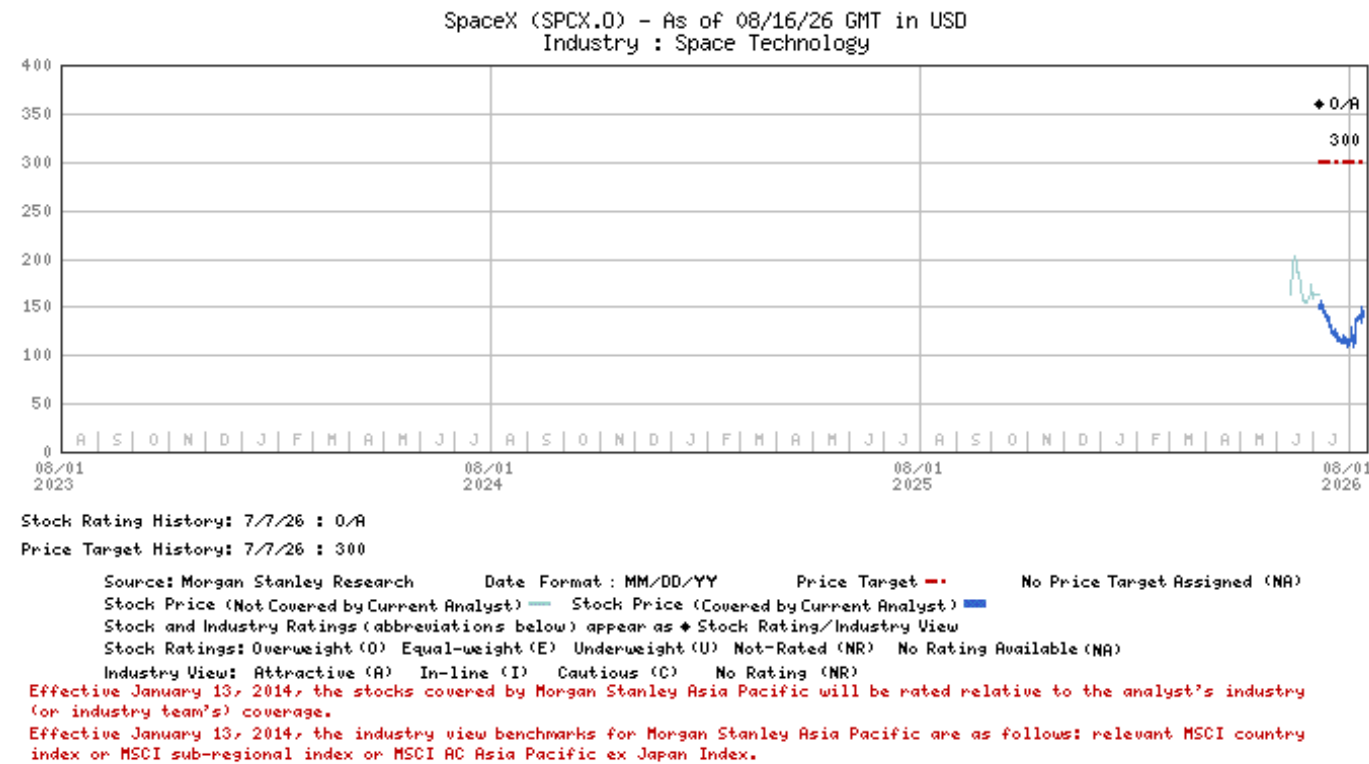
Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)



Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to <https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch>.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.

Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V which is regulated by Comision Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

INDUSTRY COVERAGE: Space Technology

COMPANY (TICKER)	RATING (AS OF)	PRICE* (08/14/2026)
Adam Jonas, CFA		
SpaceX (SPCX.O)	O (07/07/2026)	\$140.00
Justin M Lang		

Gogo Inc (GOGO.O)	E (08/14/2025)	\$2.83
Iridium Communications Inc (IRDM.O)	E (01/16/2026)	\$50.20
MDA Space Ltd (MDA.TO)	O (01/16/2026)	C\$48.86
Viasat Inc (VSAT.O)	E (12/15/2017)	\$82.87
Kristine T Liwag		
Firefly Aerospace Inc (FLY.O)	E (09/02/2025)	\$26.67
HawkEye 360 Inc (HAWK.N)	O (06/01/2026)	\$22.78
Planet Labs PBC (PL.N)	E (01/22/2023)	\$24.69
Rocket Lab USA Inc (RKL.B.O)	O (01/16/2026)	\$80.25
Virgin Galactic Holdings Inc (SPCE.N)	U (11/22/2023)	\$3.32
Voyager Technologies Inc (VOYG.N)	U (07/15/2026)	\$42.98

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.

© 2026 Morgan Stanley